

**SIMATS ENGINEERING
ASSESSMENT TOOL 2**

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0706

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL6)

Assessment Tool Weightage: 30%

Annexure A

Industry Problem Solving Task

Code of Conduct:

*I _____V.Lasya(192524186)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student : v.lasya

Annexure A

Industry Problem Solving Task

1. A star topology is ideal for a 20-person startup because all devices connect to a central switch, making the network easy to manage and troubleshoot. At the core sits a Layer 2/Layer 3 managed switch (such as a Cisco Catalyst or Netgear ProSafe) with at least 24 ports to accommodate all workstations, printers, and VoIP phones with room to grow. Each device connects to the switch using Cat6 UTP cables, which support Gigabit speeds up to 100 meters — more than sufficient for any typical office floor plan. The switch uplinks to a router/firewall (such as a Cisco RV series or pfSense appliance) that handles internet access, DHCP, and network address translation. A Wi-Fi access point connected to the switch extends wireless coverage for laptops and mobile devices. The key advantage of this design is fault isolation: if one workstation's cable fails, only that single node goes offline, leaving all other 19 employees unaffected. Centralised cable management also simplifies future moves, adds, and changes.

Answer:

A 20-person startup needs a network that is economical, easy to manage, reliable, and capable of supporting workstations, printers, VoIP phones, laptops, and internet access. A star topology is the most suitable choice because all end devices connect individually to a central managed Layer 2 or Layer 3 switch. A 24-port switch provides connections for the current users and leaves additional ports for future devices. The switch can be connected to a router or firewall for internet access, DHCP, and network address translation, while a wireless access point can provide connectivity for laptops and mobile devices.

Networking Devices and Cabling:

The main devices are a 24-port managed switch, router/firewall, wireless access point, workstations, printers, and VoIP phones. Cat6 UTP cables are suitable for device-to-switch connections and support Gigabit Ethernet over typical office cable runs up to 100 metres.

Reliability and Maintenance:

Each device has a dedicated connection to the central switch. Therefore, if one workstation or its cable fails, only that workstation loses connectivity. Centralised switching and cable management make troubleshooting, moves, additions, and changes easier.

Conclusion:

Star topology provides a good balance of cost, manageability, fault isolation, and future expansion for a 20-person startup. Its main weakness is dependence on the central switch, which can be reduced by using a reliable managed switch and suitable backup arrangements.

2. A bank demands near-zero downtime, making a full or partial mesh topology the right choice. In a full mesh, every core network device — routers, core switches, and servers — connects directly to every other device, so no single link or node failure can bring down the network. For a bank, a partial mesh is typically deployed at the core layer (connecting all core switches and firewalls to each other) while the distribution and access layers use redundant uplinks to at least two core switches. Dual enterprise-grade routers (such as Cisco ASR or Juniper MX series) provide BGP-based redundant internet connectivity. All links between core devices use fiber optic cables — either single-mode for long distances between buildings or multi-mode OM4 within the data center — because fiber is immune to electromagnetic interference, supports very high bandwidth, and is essential for maintaining low-latency financial transactions. Redundant power supplies, UPS systems, and RAID storage arrays further complement the mesh design. The justification is straightforward: when a link fails in a mesh, traffic automatically reroutes through alternative paths in milliseconds using protocols like OSPF or BGP, satisfying regulatory requirements for uptime and data integrity.

Answer:

A bank requires very high availability because network downtime can interrupt financial transactions and critical services. A full or partial mesh topology is appropriate for the critical core because it provides multiple communication paths and prevents a single link failure from isolating important systems.

Networking Devices and Transmission Media:

Core routers, core switches, firewalls, and servers should be interconnected through redundant links. Dual enterprise routers can provide redundant internet connectivity. Fibre-optic links are appropriate for the core because they provide high bandwidth, low latency, and immunity to

electromagnetic interference. Single-mode fibre is suitable for longer inter-building links, while multi-mode OM4 can be used for shorter data-centre connections.

Fault Tolerance:

Routing protocols such as OSPF or BGP can select alternative paths when a link fails. Redundant power supplies, UPS systems, and RAID storage further improve service availability. If one communication link becomes unavailable, traffic can continue through another available path.

Conclusion:

Mesh topology is justified for a bank's critical network because redundancy and fault tolerance are more important than minimum installation cost. A partial mesh at the core provides strong resilience while avoiding the excessive cabling and complexity of a full mesh everywhere.

3. For a small classroom lab with a tight budget, bus topology offers the lowest hardware cost because all computers share a single common communication line — the bus — eliminating the need for a central switch or hub. A single coaxial cable (RG-58, 50-ohm) or, in a modern variation, a single Cat5e/Cat6 cable daisy-chained through an inexpensive unmanaged hub runs along the length of the room. Each computer connects to the backbone using a T-connector and a BNC connector in a classic coaxial setup, with terminators (50-ohm) placed at both ends of the cable to absorb signals and prevent reflection. For a more contemporary low-cost implementation, a single inexpensive 8- or 16-port unmanaged switch replaces the terminators and uses Cat5e UTP patch cables to each workstation — achieving bus-like simplicity at minimal cost. The primary limitation is that the network is susceptible to collisions and a single cable break disrupts all connected machines, but for a supervised classroom environment with light usage (internet browsing, file sharing), this trade-off is entirely acceptable. No dedicated server is required; a teacher's PC can share resources using built-in Windows/Linux file sharing.

Answer:

A small classroom laboratory with a limited budget needs a simple network for light activities such as internet browsing and file sharing. A bus topology provides low hardware cost because devices share a common communication path and no central switch is required in the classic implementation.

Cabling and Devices:

A traditional bus can use a single 50-ohm RG-58 coaxial backbone with T-connectors, BNC connectors, and 50-ohm terminators at both ends. Each computer is attached to the backbone using a T-connector and BNC connector. The terminators absorb the signal at both ends and prevent reflections.

Limitations:

The main weakness of the true bus design is that all devices share the same communication medium. Increased traffic can cause collisions, and a break in the backbone can disrupt all connected computers. For a small supervised lab with light usage, these limitations can be acceptable when budget is the main constraint.

Conclusion:

Bus topology is suitable when simplicity and low initial cost are the main priorities. It is not preferred for large or critical networks because the shared backbone creates a major point of failure and performance limitations.

4. A large university campus serves thousands of users across multiple buildings — lecture halls, labs, libraries, dormitories, and administration — making a single topology inadequate. A hybrid topology combines the best elements of multiple designs: a fiber-based ring or mesh at the campus backbone level for redundancy, a star topology within each building for manageability, and bus or tree structures within individual labs for cost-efficiency. At the campus core, high-speed fiber optic cables (single-mode, 10 Gbps or 40 Gbps) interconnect core routers and multilayer switches using a ring or partial mesh arrangement, ensuring that no single link failure isolates an entire building. Each building has its own distribution switch connected to the campus core via dual fiber uplinks. Within each building, a traditional star topology branches from the distribution switch to floor-level access switches, which then connect individual devices via Cat6 cables. This hierarchical three-layer model — core, distribution, and access — is the industry standard for enterprise and campus networks because adding a new building simply means installing a new distribution switch and running fiber to the core, with no disruption to existing infrastructure. Wireless access points throughout the campus connect via PoE (Power over Ethernet) to access switches, eliminating the need for separate power cabling. The hybrid design thus delivers redundancy at the backbone, simplicity at the building level, and scalability campus-wide.

Answer:

A university campus has thousands of users distributed across lecture halls, laboratories, libraries, hostels, and administrative buildings. A single topology cannot efficiently satisfy redundancy, manageability, cost, and scalability requirements, so a hybrid topology is suitable.

Backbone and Building Design:

A redundant fibre-based ring or partial mesh can be used at the backbone. Each building can have a distribution switch with dual fibre uplinks to the campus core. Inside each building, a hierarchical star arrangement can connect floor-level access switches and end devices using Cat6 cabling.

Scalability and Technologies:

The core, distribution, and access layers provide a structured design that supports expansion. A new building can be added by installing a distribution switch and connecting it to the core without redesigning existing buildings. Wireless access points can connect to access switches using Power over Ethernet, reducing the need for separate power cabling.

Conclusion:

Hybrid topology is the most suitable choice for a large university campus because it combines backbone redundancy, building-level manageability, cost efficiency, and campus-wide scalability.

5. Designing a network for a company with multiple departments involves creating a structured and scalable architecture using switches and routers to ensure efficient communication and security. Each department (such as HR, Finance, Sales, and IT) is connected through dedicated Layer 2 switches that form the access layer, allowing devices like computers and printers to communicate within the same department. These switches are then connected to a central Layer 3 switch or router, which acts as the core layer and enables communication between different departments through inter-VLAN routing. Virtual LANs (VLANs) are configured to logically separate each department's network, improving security and reducing broadcast traffic. A router is also used to connect the internal network to external networks such as the internet, often with firewall functionality for security. Redundant links and backup paths can be included to ensure reliability, while proper IP addressing and subnetting are applied for efficient network management. This

hierarchical design improves performance, enhances security, and allows easy expansion as the company grows.

Answer:

A company with departments such as HR, Finance, Sales, and IT requires efficient communication while keeping departmental traffic logically separated. A hierarchical network using access switches and a central Layer 3 switch or router is suitable.

VLAN and Routing Design:

Each department can have a dedicated Layer 2 access switch. VLANs can logically separate HR, Finance, Sales, and IT even when they share the same physical infrastructure. The Layer 3 switch or router performs inter-VLAN routing when communication between departments is required.

Security and Reliability:

The internal network connects to an external router or firewall for internet access and security controls. VLAN separation reduces broadcast traffic and provides a logical security boundary between departments. Redundant links and backup paths can be included for important connections.

IP Addressing and Scalability:

Proper IP addressing and subnetting should be planned for each VLAN so that departments have clearly defined address ranges. New users and departments can be added by extending the access layer and assigning the required VLAN and IP subnet.

Conclusion:

The hierarchical VLAN-based design provides a practical combination of performance, security, manageability, reliability, and scalability. It is well suited to a growing company.

RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Problem Analysis	15%	Clearly identifies and deeply analyzes the industry problem with all factors	Good understanding with minor gaps	Basic understanding; some key aspects missing	Poor or incorrect understanding of the problem
Solution Design	20%	Innovative, practical, and industry-relevant solution	Effective solution with minor limitations	Partially suitable solution	Inappropriate or unclear solution
Technical Implementation	20%	Fully functional, accurate, and well-executed implementation	Mostly functional with minor errors	Partially implemented solution	Incomplete or non-functional implementation
Use of Tools & Technologies	10%	Appropriate and advanced use of relevant tools and technologies	Correct use of tools	Limited or basic use of tools	Incorrect or no use of tools
Problem-Solving Approach	10%	Logical, systematic, and well-structured approach	Mostly logical approach	Somewhat disorganized approach	No clear approach
Innovation & Creativity	10%	Highly creative and original solution	Some creativity	Limited originality	No creativity
Testing & Validation	5%	Thorough testing with real-world scenarios	Adequate testing	Minimal testing	No testing
Documentation & Presentation	5%	Clear, professional, and well-documented	Good documentation	Basic documentation	Poor or no documentation
Teamwork / Collaboration	5%	Excellent coordination and contribution (if team-based)	Good collaboration	Limited participation	No collaboration or contribution